Ramkumar Natarajan

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Education

Carnegie Mellon University — Pittsburgh, PA
PhD in Robotics — *Aug 2018 – Ongoing*
Advisors: *Maxim Likhachev and Howie Choset*

Worcester Polytechnic Institute — Worcester, MA
MS in Robotics (Ranked 1st in class) — *Aug 2015 – June 2017*

SASTRA Unviersity — Thanjavur, India
BS in Electrical Engineering — *Aug 2011 – May 2015*

Work Experience

Graduate Research Assistant — Aug 2018 – Present
Search Based Planning Lab and Biorobotics Lab at The Robotics Institute, CMU — *Pittsburgh, PA*

- PhD candidate developing motion planning algorithms for highly dynamic tasks

Senior Robotics Researcher — Jun 2017 – Jul 2019
RobotWits LLC (Acquired by Waymo) — *Pittsburgh, PA*

- As a part of an early startup, I was solely responsible for the company's entire technical development for months together. I established the operational procedures, development processes, and the hiring pipeline. Subsequently, I facilitated the hiring of four new engineers and mentored them on system design and development.
- Led the design and development of the multi-tier motion planning stack containing low-level high-speed planning, high-level route planning, and deliberative planning under uncertainty for seamless integration with perception, mapping, and control subsystem as well as the simulator through a middleware.
- Implemented the core algorithmic components of motion planning stack while maintaining soft real-time constraints on the application layer.
- Led the integration of the motion planning stack with other autonomy subsystems on physical vehicles for various clients and delivered multiple projects from concept to completion.
- Designed the fail-operational architecture for redundancy in the planning subsystem of self-driving vehicles.

Robotics Engineer — May 2016 – May 2017
Bossanova Robotics — *Pittsburgh, PA*

- Extended Viola Jones rapid object detection algorithm to multi-channel images (color, depth, and intensity).
- Incorporated graph reductionist and tree decomposition methods to parallelize filtering and smoothing for SLAM.

Internships

Research Intern — May 2021 – Aug 2021
Near Earth Autonomy — *Pittsburgh, PA*

- Planning under uncertainty for autonomous landing of an unmanned aerial vehicle on a naval destroyer

Visiting Researcher — Aug. 2014 – May 2015
Indian Institute of Technology(IIT) Madras — *Chennai, India*

- Real-time navigation for Pioneer P3-DX mobile robot using lifelong SLAM

Publications

[1] Ramkumar Natarajan, Chaoqi Liu, Howie Choset, and Maxim Likhachev. **Implicit Graph Search for Planning on Graphs of Convex Sets**. In *Robotics: Science and Systems (RSS)*, 2024.

[2] Ramkumar Natarajan, Shohin Mukherjee, Howie Choset, and Maxim Likhachev. **PINSAT: Parallelized Interleaving of Graph Search and Trajectory Optimization for Kinodynamic Motion Planning**. *Under Review at IROS 2024 - arXiv preprint arXiv:2401.08948*, 2024.

[3] Ramkumar Natarajan, Garrison L. H. Johnston, Nabil Simaan, Maxim Likhachev, and Howie Choset. **Long Horizon Planning through Contact using Discrete Search and Continuous Optimization**. *Under Review at IEEE Transactions on Robotics.*

[4] Ramkumar Natarajan*, Hanlan Yang*, Qintong Xie, Yash Oza, Manash Pratim Das, Fahad Islam, Muhammad Suhail Saleem, Howie Choset, and Maxim Likhachev. **Preprocessing-based Kinodynamic Motion Planning Framework for Intercepting Projectiles using a Robot Manipulator**. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024.

[5] Ramkumar Natarajan, Garrison LH Johnston, Nabil Simaan, Maxim Likhachev, and Howie Choset. **Torque-limited Manipulation Planning through Contact by Interleaving Graph Search and Trajectory Optimization**. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 8148–8154. IEEE, 2023.

[6] Ramkumar Natarajan, Howie Choset, and Maxim Likhachev. **Interleaving Graph Search and Trajectory Optimization for Aggressive Quadrotor Flight**. *IEEE Robotics and Automation Letters*, 6(3):5357–5364, 2021.

[7] Ramkumar Natarajan, Muhammad Saleem, Sandip Aine, Maxim Likhachev, and Howie Choset. **A-MHA*: Anytime Multi-Heuristic A***. In *Proceedings of the International Symposium on Combinatorial Search*, volume 10, pages 192–193, 2019.

[8] Ramkumar Natarajan and Michael A Gennert. **Efficient Factor Graph Fusion for Multi-robot Mapping and Beyond**. In *2018 21st International Conference on Information Fusion (FUSION)*, pages 1137–1145. IEEE, 2018.

[9] Siddharthan Rajasekaran*, Ramkumar Natarajan*, and Jonathan D Taylor. **Towards Planning and Control of Hybrid Systems with Limit Cycle using LQR Trees**. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5196–5203. IEEE, 2017.

Patents

[1] Jonathan Davis Taylor and Ramkumar Natarajan. **Color Haar Classifier for Retail Shelf Label Detection**. Google Patents, June 13 2019. US Patent App. 16/219,238.

Technical Reports

[1] Sri Ramana Sekharan, Ramkumar Natarajan, and Siddharthan Rajasekaran. **Transfer from multiple linear predictive state representations (PSR)**. *arXiv preprint arXiv:1702.02184*, 2017.

Thesis

[1] Ramkumar Natarajan. **Efficient Factor Graph Fusion for Multi-robot Mapping**, 2017.

Technical Skills

Languages: C/C++, Python, Matlab, Julia, Verilog
Developer Tools: ROS, Eigen, Boost, PyTorch, OpenCV, Git, Docker, SolidWorks, Arduino

Invited Talks

Implicit Graph Search for Planning on Graphs of Convex Sets — Nov 2023
Robot Locomotion Group, MIT — *Cambridge, MA*

Use of Topology in Optimal Motion Planning — ICRA 2019
Workshop on Topological Methods in Motion Planning — *Montreal, Canada*

Teaching

16-745 Optimal Control and Reinforcement Learning Spring 2021
Teaching Assistant for course taught by Chris Atkeson *The Robotics Institute, CMU*

16-782 Planning and Decision-making in Robotics Fall 2020
Teaching Assistant for course taught by Maxim Likhachev *The Robotics Institute, CMU*

ECE 3829: Advanced Digital System Design using FPGAs Spring 2016
Teaching Assistant *WPI*

Scholastic Achievements

Ranked 1^{st} in the MS in Robotics Engineering in a class of 48 students. 2017
Worcester Polytechnic Institute (WPI) *Worcester, MA*

Junior Research Fellowship 2013
Center for Artificial Intelligence & Robotics - Defense Research and Development Organization (CAIR - DRDO) *India*

References

Maxim Likhachev Professor, The Robotics Institute, CMU
maxim@cs.cmu.edu

Howie Choset Professor, The Robotics Institute, CMU
choset@cs.cmu.edu